Toric cohomological rigidity of simple polytopes

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This presentation is based on the work jointly with

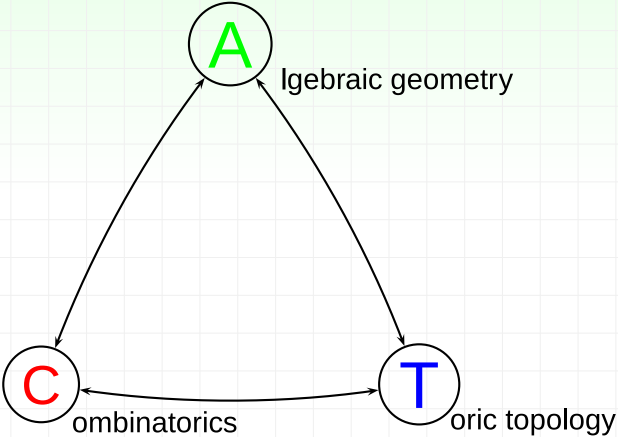
Taras Panov (Moscow State Univ.)

Dong Youp Suh (KAIST)

You can see the paper on the web arXiv:0807.4800.

Toric topology

This work is located



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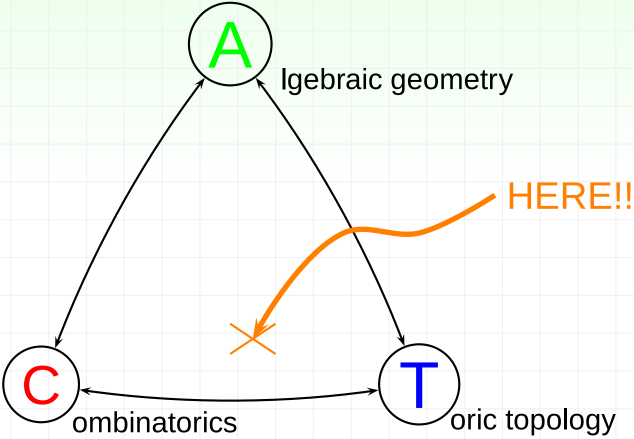
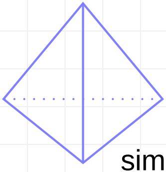


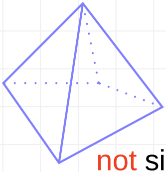
Simple polytope

A (convex) polytope is the convex hull of a finite set of points in some $\mathbb{R}^n$. Two polytopes are combinatorially equivalent if there is a bijection between their sets of faces that preserves the inclusion relation.

An n-dim. polytope is simple if each vertex is the intersection of exactly n facets, i.e. faces of dim. $n - 1$.



simple



not simple

Toric Manifold

- Toric Variety : a normal complex algebraic variety with $(\mathbb{C}^*)^n$ action having a dense orbit.
- Toric manifold : a compact non-singular toric variety.

We regard the compact torus T^n as the standard subgroup in $(\mathbb{C}^*)^n$, i.e., $T^n = \{z \in (\mathbb{C}^*)^n : |z| = 1\} \cong (S^1)^n$.

1. The action of T^n on a toric manifold is *locally standard*.

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The *standard* action T^n on $\mathbb{C}^n$
$(t_1, \ldots, t_n) \cdot (z_1, \ldots, z_n) = (t_1 z_1, \ldots, t_n z_n)$. The orbit space of this action is the positive cone $\mathbb{R}^n_+$.
Globally, the orbit space for a locally standard T^n actions on M^{2n} is an n-dimensional manifold with corners.

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1. The action of T^n on a toric manifold is *locally standard*.

2. The orbit space of a toric manifold with T^n can be identified with the simple polytope

Quasitoric manifold

By Davis and Januszkiewicz, we have the notion of a topological generalization by taking these two properties

- Quasitoric Manifold : a closed smooth manifold M of dim. $2n$ with a smooth $(S^1)^n$ action such that
  - the action is *locally standard*,
  - the orbit space $M/(S^1)^n$ is a simple convex polytope.

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Example

- $\mathbb{C}P^n$ with the standard T^n-action

$$(t_1, \dots, t_n) \cdot [z_1 : \cdots : z_n : z_{n+1}] = [t_1 z_1 : \cdots : t_n z_n : z_{n+1}]$$

  is a quasitoric manifold over the n-simplex Δ^n.
- $\prod \mathbb{C}P^{n_i}$ is a quasitoric manifold over $\prod \Delta^{n_i}$.

Cohomological rigidity of simple polytope

- $M \to P, N \to Q$: quasitoric manifolds

When does $H^*(M, \mathbb{Z}) \cong H^*(N, \mathbb{Z})$ imply $P \approx Q$?

P is (cohomologically) rigid if whenever there is another polytope Q with $H^*(M, \mathbb{Z}) \cong H^*(N, \mathbb{Z})$, there is a combinatorial equivalence $P \approx Q$.

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Which polytopes are rigid?

Toric topology and Combinatorics

In the general topology,

1. the information of Lie-group action on M is dependent on the symmetry of M, that is, orbit space.
2. cohomology ring of M is defined independently the Lie-group action on M.

But!

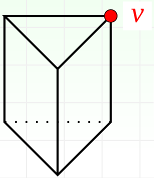
If P is rigid, then the combinatorial structure of P is determined by the cohomology ring of supporting quasitoric manifold M.

Example of non-rigid simple polytope

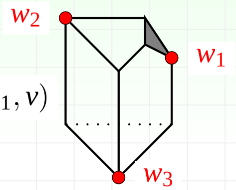
Not all polytopes are rigid:

- $vc(P,v)$ = the vertex cut of P at v.

P_1



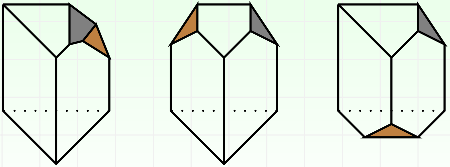
$P_2 := vc(P_1, v)$



$$\begin{array}{c} M_1 = \mathbb{C}P^2 \times \mathbb{C}P^1 \\ \Big\downarrow \pi \\ P_1 \end{array} \qquad \begin{array}{c} M_2 = \text{ blow up of } M_1 \text{ at } \pi^{-1}(v) \\ \Big\downarrow \pi' \\ P_2 \end{array}$$

Example of non-rigid simple polytope

$$P_{3,i} := vc(P_2, w_i)$$



$$M_{3,i} = \text{ blow up of } M_2 \text{ at } \pi'^{-1}(w_i)$$

$$\Big\downarrow$$

$$P_{3,i}$$

$$M_{3,i} \cong (\mathbb{C}P^2 \times \mathbb{C}P^1)\sharp\mathbb{C}P^3\sharp\mathbb{C}P^3$$

History

M.Davis and T.Januszkiewicz (1991)

$$\beta^{2i}(H^*(M)) = h_i(P)$$

Cohomology determines *face numbers* of P.

M.Masuda, T.Panov (2007)

- every hypercube I^n.

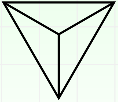
Rigid polytopes

The following polytopes are rigid ;

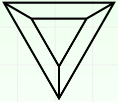
- every 2-dim polytope, i.e., polygon,
- every triangle-free n-dim simple polytope with facet numbers$\leq 2n+2$,
- any product of simplices $\prod_{i=1}^{k} \Delta^{n_i}$,
- vertex cut of a product of simplices $vc(\prod_{i=1}^{k} \Delta^{n_i})$,
- Dodecahedron.

Rigid polytopes

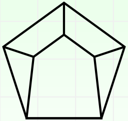
Rigid 3-polytopes up to 9 facets.



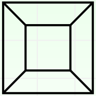
3^4



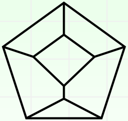
$3^2 4^3$



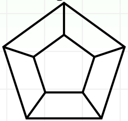
$3^2 4^2 5^2$



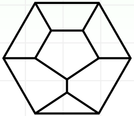
4^6



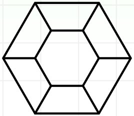
$3^1 4^3 5^3$



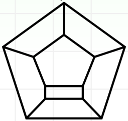
$4^5 5^2$



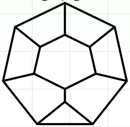
$3^1 4^3 5^3 6^1$



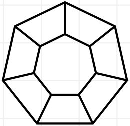
$4^6 6^2$



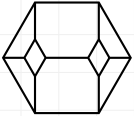
$4^4 5^4$



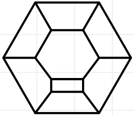
$3^1 4^4 5^2 6^1 7^1 (ii)$



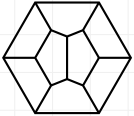
$4^7 7^2$



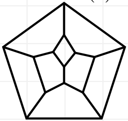
$4^6 6^3$



$4^5 5^2 6^2$



$4^4 5^4 6^1$



$4^3 5^6$

Corollary

S.Choi, M.Masuda and D.Y.Suh (2008)

- M : quasitoric manifold over $\prod_{j=1}^{m} \Delta^{n_j}$.

$$H^*(M) \cong H^*\left(\prod_{j=1}^{m} \mathbb{C}P^{n_j}\right) \Longrightarrow M \underset{\text{homeo.}}{\approx} \prod_{j=1}^{m} \mathbb{C}P^{n_j}$$

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Face ring of simple polytopes

- P : n-simple polytope with m facets
- $\mathfrak{F}(P) = \{F_1, \ldots, F_m\}$: set of facets

The face ring (or the Stanley-Reisner ring) of P is the quotient ring

$$\mathbb{Q}(P) = \mathbb{Q}[v_1, \ldots, v_m]/\mathcal{I}_P,$$

where $\mathcal{I}_P$ is the homogeneous ideal generated by all square-free monomials $v_{i_1} \cdots v_{i_s}$ such that $\bigcap_{\ell=1}^{s} F_{i_\ell} = \emptyset$.

W.Bruns, J.Gubeladze (1996)
Let P and Q be two n-simple polytopes.

$$\mathbb{Q}(P) \cong \mathbb{Q}(Q) \text{ as } \mathbb{Q}\text{-algebras} \iff P \approx Q$$

Cohomology of quasitoric manifold M over P

- $\mathfrak{F} = \{F_1, \ldots, F_m\}$: the set of facets of P

(known) $H^*_T(M) := \mathbb{Z}[v_1, \ldots, v_m]/\mathcal{I}_P \cong \mathbb{Z}(P)$: face ring.

(known) $H^*(M) := H^*_T(M)/J \cong \mathbb{Z}(P)/J$ as rings,

where J is the ideal generated by some linear combination of $v_1, \ldots, v_m$.

Tor-algebra

- A : finitely generated graded algebra
- M, N : finitely generated A-modules

A free resolution $[R, d]$ of M is an exact sequence

$$0 \longrightarrow R^{-n} \xrightarrow{d} R^{-n+1} \xrightarrow{d} \cdots \xrightarrow{d} R^0 \xrightarrow{d} M \longrightarrow 0\ ,$$

Take $\otimes_A N$ to get

$$0 \longrightarrow R^{-n} \otimes_A N \xrightarrow{d\otimes 1} R^{-n+1} \otimes_A N \xrightarrow{d\otimes 1} \cdots \xrightarrow{d\otimes 1} R^0 \otimes_A N \longrightarrow 0$$

Then $\mathrm{Tor}_A^{-i}(M, N)$ is defined to be

$$\mathrm{Tor}_A^{-i}(M, N) := H^{-i}(R^{-*} \otimes_A N).$$

Since everything is graded,

$$\mathrm{Tor}_A^{-i}(M, N) = \bigoplus_j \mathrm{Tor}_A^{-i,j}(M, N)$$

Tor-algebra of simple polytope

- $A := \mathbb{Q}[v_1, \ldots, v_m]$: the polynomial graded ring

Note that both $\mathbb{Q}(P)$ and $\mathbb{Q}$ are graded A-modules. Then we can define $\mathrm{Tor}_A(\mathbb{Q}(P), \mathbb{Q})$.
The bigraded Betti number $\beta^{-i,j}(P)$ of P is defined by

$$\beta^{-i,j}(P) = \dim_{\mathbb{Q}} \mathrm{Tor}_A^{-i,j}(\mathbb{Q}(P), \mathbb{Q}).$$

Bigraded Betti numbers are combinatorial invariants of P!

Theorem

- M, N : two quasitoric manifolds over P and Q, resp.

$$H^*(M) \cong H^*(N) \Longrightarrow \mathrm{Tor}(\mathbb{Q}(P), \mathbb{Q}) \cong \mathrm{Tor}(\mathbb{Q}(Q), \mathbb{Q})$$
$$\Longrightarrow \beta^{-i,j}(P) = \beta^{-i,j}(Q)$$

To show uniqueness of the bigraded Betti number of P is one of the ways to decide the rigidity of P.

Sketch of Proof - a product of simplices

- $\sigma(P) = \sum_{j \geq 2} j \beta^{-1,j}$

$\sigma(P)$ is nothing but the sum of the degrees of all elements of a minimal basis of I_P.

[Lemma]

$$\sigma(P) = m \Longleftrightarrow P = \prod \Delta^{n_i}$$

(Proof of lemma)

$$\begin{aligned} \mathbb{Q}(P) &\cong \mathbb{Q}[x_1, \ldots, x_m]/I_P \\ &\cong \otimes_{i=1}^{t} \mathbb{Q}[y_{i,0}, \ldots, y_{i,n_i}]/(g_i) \\ &\cong \otimes_{i=1}^{t} \mathbb{Q}(\Delta^{n_i}) \cong \mathbb{Q}\left(\prod_{i=1}^{t} \Delta^{n_i}\right). \end{aligned}$$

$\square$

Thank you for listening!